

During today's lab the most surprising discovery for me was how images are represented by a stack of numbers, those stacks being 3D matrices. While I initially expected computers to recognize a vast amount of colors directly. I learned that digital images are broken down into three colors, red, green, and blue. Those hues of red, green, and blue are then determined by values from 0-255. The mathematical operations we implemented in this lab relate to visual effect because all of numbers make up the images.

The most challenging technique from today's lab for me was Histogram Analysis and Enhancement. As I struggled to understand what graphs had to do with digital images. But after reading that portion of the lab into fuller detail, I realized that the histograms count the pixels intensity. The most difficult part was understanding how this could fix images by way of Histogram Equalization. I realized that the computer looks at the values to determine how to contrast the image.

Today's lab connects to the Nano Banana demonstration because all the operations we performed were a manual application of a filter. Where Nano Banana differs from my actions in the lab is that it learns filters from data and can carry them out on its own. I can envision social media applications utilizing these techniques for filters, such as Snapchat, Instagram, and TikTok. How I would combine traditional and AI approaches for future projects would be to use AI to do the initial work, and if there was some fine tuning that requires a human touch. I would then use a traditional approach and make those changes.

The aspect of image processing that interests me the most for further exploration is being able to change the RGB into an image. Specially I'm interested in finding out more about how you can change color contrast by altering those numbers. I dove into this a little bit by carrying out my changes to the lab. For my changes to the lab, I first changed the red in the image to [75: 25, 75:25] = [255, 100, 100] and realized my changes made the red go away. I also changed the brightness to 100 and darkness to -100, and contrast to 3. I realized that by doing this it shifted the mean from 38.7 to 136.8 which shows it directly adjusted the light level. This solidified my understanding that images are numbers for computers so to alter the image you alter the numbers. I am curious as to how images are processed by computers to learn certain things like what a human or an animal is. Being that images are a series of numbers to a computer; I am curious that when AI is tasked to determine something to this effect. Is the technology looking at the shape of the 3D matrix or the pattern of numbers to differentiate?